# BIHAR ENGINEERING UNIVERSITY, PATNA

## DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

**B.Tech 4th Semester • Examination Resource**

# ELECTRONICS MATERIALS & INSTRUMENTATION

**Exam Date: Saturday, 7 Nov 2026 (2:00 PM - 5:00 PM)**

### Included Question Papers

| Year | Paper Title | Course Code | Examination Session |
|---|---|---|---|
| **2024** | Semiconductor Physics and Devices | 104405 | B.Tech 4th Semester Examination, 2024 |
| **2023** | Semiconductor Physics and Devices | 104405 | End Semester Examination - 2023 (Semester IV) |
| **2024** | Electronics Instruments and Measurement | 104604 | B.Tech 6th Semester Examination, 2024 |
| **2024** | Measurements and Instrumentation | 110601 | B.Tech 6th Semester Examination, 2024 |
| **2022** | INSTRUMENTATION AND CONTROL | 102404 | B.Tech 4th Semester Exam., 2022 |

**Apna BEU Question Bank Archive** • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

# BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

## SEMICONDUCTOR PHYSICS AND DEVICES

Time: 03 Hours | Code: 104405 | Full Marks: 70

**Instructions:**

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

**Q.1 Choose the correct option/answer the following (Any seven question only):** **[2x7=14]**

a. Which of the following best describes the energy band model of a semiconductor?

i. Electrons are localized within individual atoms
ii. Electrons occupy energy bands separated by a band gap
iii. Electrons are free to move without any restriction
iv. All energy levels are completely filled

b. In an extrinsic semiconductor at equilibrium, the Fermi level $E_F$ shifts towards:

i. The conduction band in p-type material
ii. The conduction band in n-type material
iii. The valence band in n-type material
iv. The mid-gap energy level in doped semiconductors

c. Which of the following phenomena describe the generation and recombination processes in semiconductors?

i. Band-to-band recombination
ii. Impact ionization
iii. Auger recombination
iv. All of the above

d. What is the primary cause of the built-in potential in a PN junction diode?

i. Applied external voltage across the junction
ii. Diffusion of majority carriers
iii. Recombination of electrons and holes
iv. Ionization of donor and acceptor atoms

e. Which of the following mechanisms is responsible for junction breakdown in a Zener diode?

i. Avalanche multiplication
ii. Thermal runaway
iii. Tunneling of carriers across the junction
iv. Both (i) and (iii)

f. In which mode of operation does a Bipolar Junction Transistor (BJT) act as an amplifier?

i. Cutoff mode
ii. Active mode
iii. Saturation mode
iv. Reverse-active mode

g. What is the expression for common-base current gain ($\alpha$) of a BJT?

i. $\alpha = I_C/I_B$
ii. $\alpha = I_E/I_C$
iii. $\alpha = I_C/I_E$
iv. $\alpha = I_B/I_C$

h. In a Junction Field-Effect Transistor (JFET), the drain current ($I_D$) in the saturation region is primarily controlled by:

i. The gate-to-source voltage ($V_{GS}$)
ii. The source current ($I_S$)
iii. The drain-to-source voltage ($V_{DS}$)
iv. The drain resistance

i. Which of the following correctly describes the threshold voltage ($V_{th}$) of a MOSFET?

i. The voltage at which the drain current becomes zero
ii. The voltage at which the device enters saturation
iii. The minimum gate voltage required to form an inversion layer at the semiconductor-oxide interface
iv. The voltage at which the oxide layer begins conducting current

j. Which of the following devices is primarily used for triggering other semiconductor devices like SCRs?

i. TRIAC
ii. UJT
iii. DIAC
iv. MOSFET

**Q.2 Solve both questions :** **[14]**

a. Explain the concept of the bonding model and energy band model in semiconductors.

b. Explain drift and diffusion current mechanisms in semiconductors. Derive the Einstein relation that connects diffusion coefficient and mobility.

**Q.3 Solve both questions :** **[14]**

a. Discuss the different types of recombination mechanisms and their significance in semiconductor devices.

b. Write and explain the continuity equation for charge carriers in semiconductors. Derive the minority carrier diffusion equation and discuss its importance in the operation of semiconductor devices.

**Q.4 Solve both questions :** **[14]**

a. Explain the Ideal Diode Equation. Derive it quantitatively using the band model and boundary conditions, stating the assumptions made in the derivation.

b. Explain the working principles and applications of special diodes such as the Zener diode, Tunnel diode, Varactor diode and Schottky diode. How do their characteristics differ from a conventional PN junction diode?

**Q.5 Solve this question :** **[14]**

Discuss the two primary mechanisms of junction breakdown in a PN junction diode- Avalanche breakdown and Zener breakdown. Compare their occurrence conditions and characteristics.

**Q.6 Solve this question :** **[14]**

Describe the impact of base width modulation (Early effect) on BJT operation. How does it affect the common-emitter current gain and output characteristics? Include relevant equations in your explanation.

**Q.7 Solve this question :** **[14]**

Explain the deviations from ideal MOS behaviour, including metal-semiconductor work function differences, fixed oxide charges, interface traps and threshold voltage shifts. How do these non-ideal effects influence MOSFET performance?

**Q.8 Solve this question :** **[14]**

Discuss the working principles and characteristics of different optoelectronic devices: PIN Photodiode, Avalanche Photodiode, Solar Cell and LED. Compare their applications.

**Q.9 Solve this question :** **[14]**

Explain the working of a solid-state laser diode. How does it differ from an LED in terms of construction, working principle, and applications? Discuss its advantages in optical communication.

# BIHAR ENGINEERING UNIVERSITY, PATNA

## End Semester Examination - 2023 (Semester IV)

## SEMICONDUCTOR PHYSICS AND DEVICES

Time: 03 Hours | Code: 104405 | Full Marks: 70

**Instructions:**

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

**Q.1 Write the short answer of the following (Any seven question only):** **[2x7=14]**

a. Define the term drift velocity.

b. Define Fermi energy level.

c. Define doping process.

d. Why is the width of the base region kept very small as compared to other regions in BJT's?

e. Write the difference between Zener and Avalanche breakdown.

f. What do you mean by solid state?

g. Define the effective mass and express it in terms $(E, K)$.

h. Define conductivity and mobility.

i. Write two differences between BJT and FET.

j. Define the varactor diode.

**Q.2 Solve both questions :** **[14]**

a. Explain (i) Solar Cell (ii) LED (iii) LASER diode.

b. Discuss the working principle of SCR. How it is different from UJT?

**Q.3 Solve both questions :** **[14]**

a. What is MOS Capacitor? Draw its C-V Characteristic.

b. Explain the operation of enhancement mode MOSFET with neat diagram and compare it with JFET.

**Q.4 Solve both questions :** **[14]**

a. Discuss the punch-through breakdown in BJT. Write the expression for the punch though voltage.

b. Distinguish between the different types of transistor configurations with necessary circuit diagrams.

**Q.5 Solve both questions :** **[14]**

a. With a neat diagrams explain the working of a PN junction diode in forward bias and reverse bias and plot the V-I characteristics.

b. Discuss about the effects of temperature on PN junction diode's Forward and Reverse characteristics.

**Q.6 Solve both questions :** **[14]**

a. What is Zener diode. Explain V-I characteristics of Zener diode.

b. When a transistor is used as a switch, in which region of output characteristics it is operated? Discuss with reason.

**Q.7 Solve this question :** **[14]**

What is intrinsic semiconductor and explain the formation extrinsic semiconductors through doping?

**Q.8 Solve both questions :** **[14]**

a. Derive the expression for current generated due to drifting of charge carriers in semiconductors in the presence of electric field.

b. Derive Einstein's relation in semiconductors?

**Q.9 Write Short notes ver any two of the following:** **[7x2=14]**

a. charge carriers generation and recombination

b. UJT relaxation Oscillator

c. Schottky diode

# BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 6th Semester Examination, 2024

## ELECTRONICS INSTRUMENTS AND MEASUREMENT

Time: 03 Hours | Code: 104604 | Full Marks: 70

**Instructions:**

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

**Q.1 Choose the correct answer of the following (Any seven question only):** **[14]**

1. The high torque to weight ratio in an analog indicating instrument indicates
   - i. High Friction loss
   - ii. Low friction loss
   - iii. Nothing as regards friction
   - iv. None of the above

2. Which of the following is the most popular method for measuring low resistance?
   - i. ducter ohmmeter method
   - ii. kelvin double bridge method
   - iii. ammeter-voltmeter method
   - iv. potentiometer method

3. Which of the following method is used for the measurement of Medium Resistance?
   - i. Direct-Deflection method
   - ii. Anderson Bridge
   - iii. Carey-Foster bridge method
   - iv. Kelvin's double bridge method

4. Which of the following is a balance equation for computing the resistance?
   - i. $R_1 = R_2/R_4$
   - ii. $R_1 = R_3/R_4$
   - iii. $R_1 = R_2R_3$
   - iv. $R_1 = R_2R_3/R_4$

5. A Schering bridge can be used for
   - i. measuring voltages
   - ii. testing capacitors
   - iii. protecting the circuit from temperature rises
   - iv. measuring currents

6. What is the dependence of frequency on the balance equation?
   - i. varies by a factor of 2
   - ii. depends on the detector used
   - iii. independent
   - iv. depends on the supply magnitude

7. Which of the following can be measured using Maxwell's Inductance Bridge?
   - i. capacitance
   - ii. frequency
   - iii. mutual Induction
   - iv. Induction

8. The windings of a C.T. are
   - i. tied together
   - ii. Shorted
   - iii. wound over one another
   - iv. Grounded

9. Which of the following device is used to measure power in A.C. circuits?
   - i. ammeter
   - ii. Wattmeter
   - iii. voltmeter
   - iv. ohmmeter

10. When the moving coil in a Dynamometer type wattmeter deflects
    - i. pointer doesn't move
    - ii. current flows
    - iii. voltage is generated
    - iv. pointer moves

**Q.2 Solve both questions :** **[14]**

1. Explain how the unknown frequency is measured using Wien bridge method?

2. What is the need for standards of measurements? How are they classified?

**Q.3 Solve both questions :** **[14]**

1. How do you measure the unknown inductance using Hay's Bridge?

2. State the purpose of calibration of measuring instruments. Explain the procedure of calibration of D.C. Voltmeter by using D.C. Potentiometer.

**Q.4 Solve both questions :** **[14]**

1. How do you measure the unknown capacitance using Schering Bridge? Also draw the phasor diagram.

2. Draw the possible methods of connecting the pressure coil of a wattmeter and compare the errors. Explain the meaning of 'compensating winding' in a wattmeter and show how they help to reduce the error.

**Q.5 Solve both questions :** **[14]**

1. List out comparisons between CT's and PT's (Any four).

2. A current transformer with a bar primary has 300 turns in its secondary. Resistance and reactance of the secondary circuit are 1.5 Ω and 1.0 Ω respectively. Including the transformer winding with 5 A flowing in the secondary winding, find the phase angle error. (Assuming magnetizing mmf to be 100 AT and core loss to be 1.5 W).

**Q.6 Solve both questions :** **[14]**

1. Describe the construction and principle of operation of D' Arsonval type Galvanometer.

2. Write short notes on (a) Kelvin's double bridge (b) Digital frequency meter.

**Q.7 Solve both questions :** **[14]**

1. Describe the measurement of frequency, phase angle and time delay using oscilloscope with suitable diagrams and mathematical expressions.

2. A moving coil instrument gives a full scale deflection of 10mA, when the potential difference across its terminal is 100mV. Calculate (1) The shunt resistance for a full scale deflection corresponding to 100A (2) The resistance for full scale reading with 1000V. Calculate the power dissipation in each case?

**Q.8 Solve both questions :** **[14]**

1. Draw the block diagram of an electronic voltmeter and explain its operation.

2. Explain BH curve with a neat diagram. What is Hysteresis loop?

**Q.9 Solve both questions :** **[14]**

1. Prove that for electrodynamometer type wattmeter true power = $\frac{\cos\Phi}{\cos\Phi\cos(\Phi-\beta)} \times \textbf{actual wattmeter reading}$.
   Where $\cos\phi$ = power factor of the circuit, $\beta = \tan^{-1}(wL/R)$ where L and R are the inductance and resistance of the pressure coil.

2. Explain with a neat sketch, the construction and working principle of Megger.

# BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 6th Semester Examination, 2024

## MEASUREMENTS AND INSTRUMENTATION

Time: 03 Hours | Code: 110601 | Full Marks: 70

**Instructions:**

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

**Q.1 Choose the correct answer the following (Any seven question only):** **[14]**

1. Which system analyzes brain wave activity?
   i. EOG
   ii. EMG
   iii. ECG
   iv. EEG

2. Which type of sensor is used to measure flow?
   i. Potentiometer
   ii. Venturi tube
   iii. Hall effect sensor
   iv. Thermocouple

3. What does "Six Sigma" focus on?
   i. Enhancing resolution
   ii. Improving precision
   iii. Reducing defects in processes
   iv. Minimizing systematic errors

4. What is the mean of the following data set: 4, 6, 8, 10?
   i. 6
   ii. 2
   iii. 8
   iv. 9

5. Which instrument is used for ultrasonic diagnostic measurement?
   i. Thermocouple
   ii. Oscilloscope
   iii. Ultrasound transducer
   iv. Megger

6. What is an example of a non-invasive Cardiac output measurement diagnosis?
   i. Temperature measurement
   ii. Blood pressure measurement
   iii. ECG
   iv. All of the above

7. What is the function of a transducer in bio-medical instruments?
   i. Convert electrical signals into light
   ii. Measure electric current
   iii. Measure pressure
   iv. Convert biological signals into electrical signals

8. What does a Megger measure?
   i. Current
   ii. Insulation resistance
   iii. Capacitance
   iv. Voltage

9. What does a digital multi-meter measure?
   i. Only current
   ii. Only resistance
   iii. Only voltage
   iv. Voltage, current, and resistance

10. What type of instrument is used to measure inductance?
    i. Ohmmeter
    ii. LCR meter
    iii. Ammeter
    iv. Voltmeter

**Q.2 Solve both questions:** **[14]**

1. Discuss the use of instrument transformers for current and voltage measurement in power systems.
2. Explain the working principle of Maxwell's bridge with phasor diagram.

**Q.3 Solve both questions:** **[14]**

1. Explain the Hall effect and its application in Hall sensors for current and voltage measurements.
2. Describe the working principle of a Digital Storage Oscilloscope (DSO).

**Q.4 Solve both questions:** **[14]**

1. Define and differentiate between true value, accuracy, and precision in measurement systems.
2. Explain the concept of True RMS measurement. Why is it important for accurate AC voltage and current measurements?

**Q.5 Solve both questions:** **[14]**

1. Explain the Six-Sigma approach to quality control in industrial processes.
2. Describe the working principles of temperature transducers thermocouples and RTD.

**Q.6 Solve both questions:** **[14]**

1. Explain hysteresis and dead-band in measurement systems. How do they influence the performance of transducers?
2. Explain how bio-electric signals like ECG, EEG, and EMG are generated and measured.

**Q.7 Solve both questions:** **[14]**

1. Discuss the types of errors that occur in measurements. Explain the difference between systematic and random errors with practical examples.
2. Discuss non-invasive diagnostic measurements used in healthcare.

**Q.8 Solve both questions:** **[14]**

1. Describe the role of sensitivity in measurement systems. How is sensitivity calculated, and what are the consequences of low or high sensitivity in sensors?
2. Explain the use of Meggers in insulation resistance testing. Why is insulation testing crucial in high-voltage systems?

**Q.9 Write short notes on any two of the following:** **[14]**

1. a. CAT scan techniques
   b. Cardiovascular measurements
   c. Capacitance Measurement with phasor
   d. LVDT

# BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Exam., 2022

## INSTRUMENTATION AND CONTROL

Time: 3 hours | Code: 102404 | Full Marks: 70

**Instructions:**

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

**Q.1 Choose the correct answer of the following (any seven):** **[2x7=14]**

1. Which of the following causes a controlled variable to change?
   i. Disturbance
   ii. Change in set point
   iii. Variation in load demand
   iv. All of the above

2. What is the resolution of a DAC with a maximum voltage of 15 volts and five input lines?
   i. 0.4687
   ii. 0.4838
   iii. 0.2343
   iv. 0.2419

3. The derivative function gives the actuator _____.
   i. boost action
   ii. braking action
   iii. Both (i) and (ii)
   iv. None of the above

4. A square wave that is 20 volts at its On state and 0 volt at its Off state will produce an average DC voltage of _____ when duty cycle is 75%.
   i. 15 V
   ii. 10 V
   iii. 7.5 V
   iv. 5 V

5. The resistance of an RTD is:
   i. directly proportional to the temperature
   ii. inversely proportional to the temperature
   iii. directly proportional to the square of temperature
   iv. inversely proportional to the square of temperature

6. Which of the following valves has the greatest flow capacity of all control valves?
   i. Single-seated
   ii. Three-way
   iii. Butterfly
   iv. Ball

7. A unity feedback system has open-loop transfer function as $G(s) = \frac{16}{s(s+16)}$. The natural frequency of the system is:
   i. 16
   ii. 8
   iii. 2
   iv. 4

8. Which of the following represents the application of inductive transducers?
   i. Displacement measurement
   ii. Thickness measurement
   iii. Both displacement and thickness measurements
   iv. None of the above

9. Which of the following can act as a comparator?
   i. Op-amp with negative feedback
   ii. Op-amp with positive feedback
   iii. Op-amp without feedback
   iv. None of the above

10. Transfer function represents the input-output relationship of a system, with initial conditions as:
   i. zero
   ii. positive
   iii. negative
   iv. All of the above

**Q.2 Solve both questions :** **[14]**

1. Define the following: (i) Transducer (ii) Resolution (iii) Linearity (iv) Accuracy (v) Repeatability (vi) Calibration (vii) Backlash

2. Discuss in detail about the classification and elimination of error.

**Q.3 Solve both questions :** **[14]**

1. Differentiate among piezoelectric, inductive and capacitance-type transducers.

2. What are pyrometers? Explain with the help of neat sketch.

**Q.4 Solve both questions :** **[14]**

1. Explain the working of electrical resistance strain gauge. Explain how the resistance change of a strain gauge is measured with the help of Wheatstone bridge. Also, explain the methods of lead wire compensation and temperature compensation in case of strain gauge.

2. Explain the five basic laws of thermocouple behaviour and how these laws are useful in designing thermocouple for temperature measurement.

**Q.5 Solve all questions :** **[14]**

1. Consider a system with controller $G_c$, reference input $R$, and disturbance input $D$ acting on the plant $\frac{1}{s(s+2)}$. If $G_c$ is a proportional controller, $G_c = K_p$, what value of $K_p$ will cause the steady-state error for a unit ramp reference input to be 0.1?

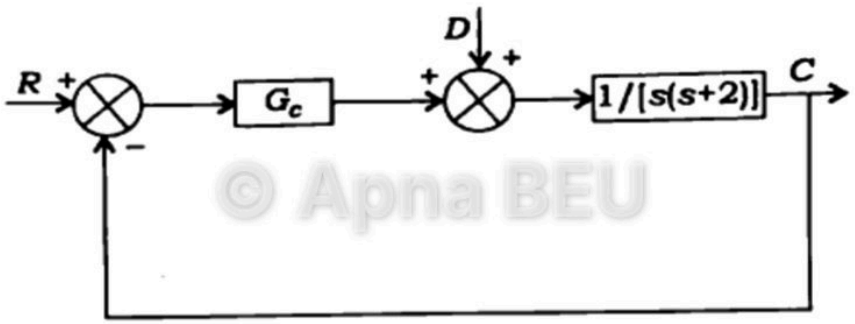
   (a) If $G_c$ is a proportional controller, $G_c = K_p$, what value of $K_p$ will cause the steady-state error for a unit ramp reference input to be 0.1?

   (b) If zero steady-state error for a unit ramp reference input is required, what form should the controller $G_c$ take?

   (c) For a unit-step disturbance input, what form of controller $G_c$ should be chosen to make the steady-state error zero?

   (d) Assume that the reference input is a unit step, the disturbance input is zero and proportional controller being used. What value (or values) of $K_p$ will yield an underdamped system with a settling time of 4 seconds?

   (e) Considering the same case as in part (d), what effect does increasing $K_p$ have on the damping ratio?



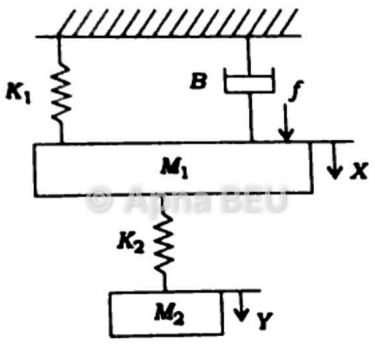
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Figure 1

**Q.6 Solve both questions :** **[14]**

1. With the help of a suitable example, explain the difference between open-loop control and closed-loop control.

2. For a given mechanical translational system with masses $M_1, M_2$, springs $K_1, K_2$, and damper $B$, find out the transfer function $\frac{X(s)}{F(s)}$.



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Figure 2

**Q.7 Solve both questions :** **[14]**

1. Explain the operations of the following: (i) Inverting op-amp (ii) Summing op-amp (iii) Differentiating op-amp (iv) Integrating op-amp.

2. What are the important requirements of a signal conditioner and how are they met?

**Q.8 Solve this question :** **[14]**

1. For the open-loop transfer function $G(s) = \frac{400(s+1)(s+10)}{s^3(s+100)(s+1000)}$, sketch the Bode magnitude and phase diagrams. Sketch the diagrams for each term and then also sketch the total diagrams. Show the slopes of each section and of the total diagram.

**Q.9 Write short notes on any four of the following:** **[14]**

1. a. Sensitivity analysis
   b. Pneumatic actuators
   c. Flow measurement sensors
   d. ADC using successive-approximation register (SAR)
   e. Optical encoders